

Multiplexer and Demultiplexer

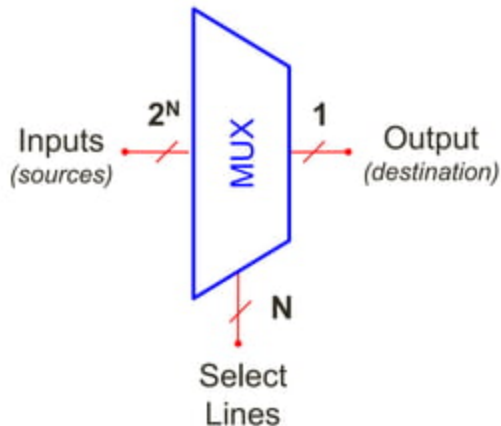


Presented by
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Multiplexer (MUX)

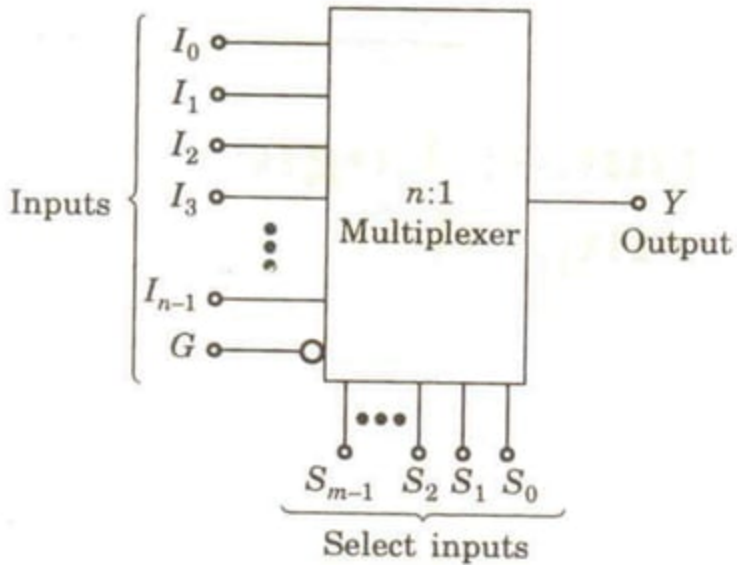
- A MUX is a digital switch that has multiple inputs (sources) and a single output (destination).
- The select lines determine which input is connected to the output.
- MUX Types
 - 2-to-1 (1 select line)
 - 4-to-1 (2 select lines)
 - 8-to-1 (3 select lines)
 - 16-to-1 (4 select lines)

Multiplexer
Block Diagram

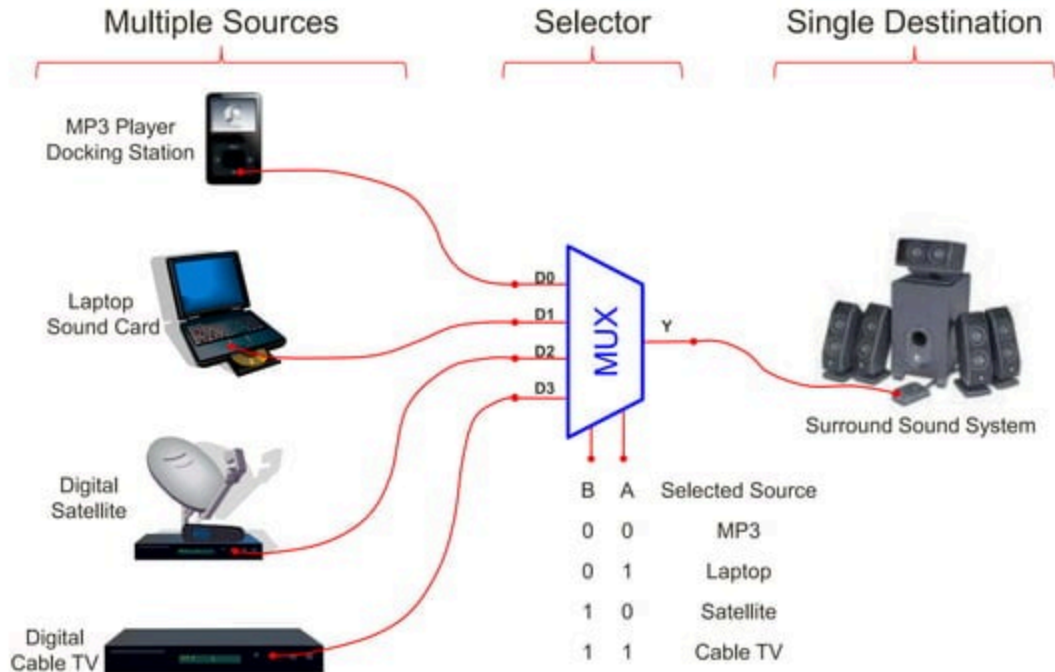


MULTIPLEXERS

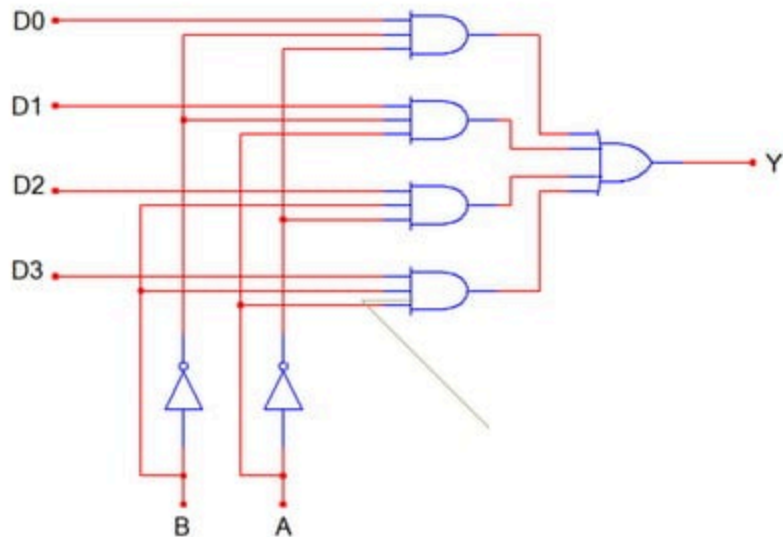
Multiplexer



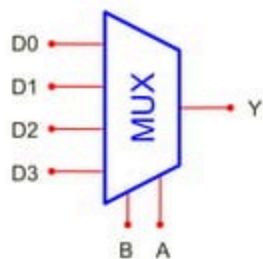
Typical Application of a MUX



4-to-1 Multiplexer (MUX)



$$Y = A'B'D0 + AB'D1 + A'BD2 + ABD3$$



B	A	Y
0	0	D0
0	1	D1
1	0	D2
1	1	D3

Truth Table of a 4:1 Multiplexer

$$Y = (\bar{S}_1 \bar{S}_0 I_0 + \bar{S}_1 S_0 I_1 + S_1 \bar{S}_0 I_2 + S_1 S_0 I_3) \cdot \bar{G}$$

Enable input G	Select inputs		Output Y
	S_1	S_0	
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	X	X	0

$S_0=0, S_1=1$

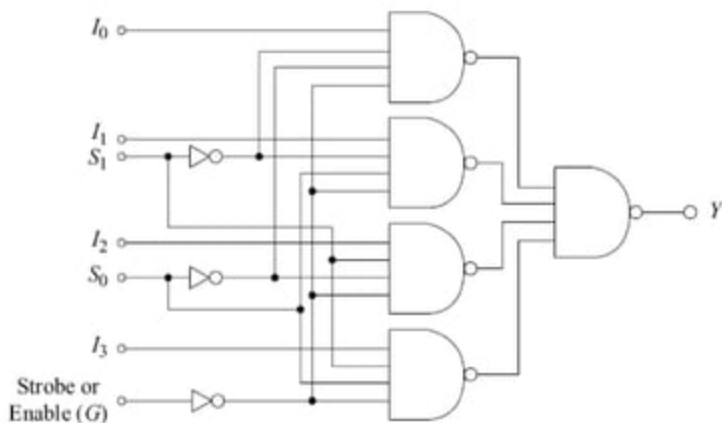
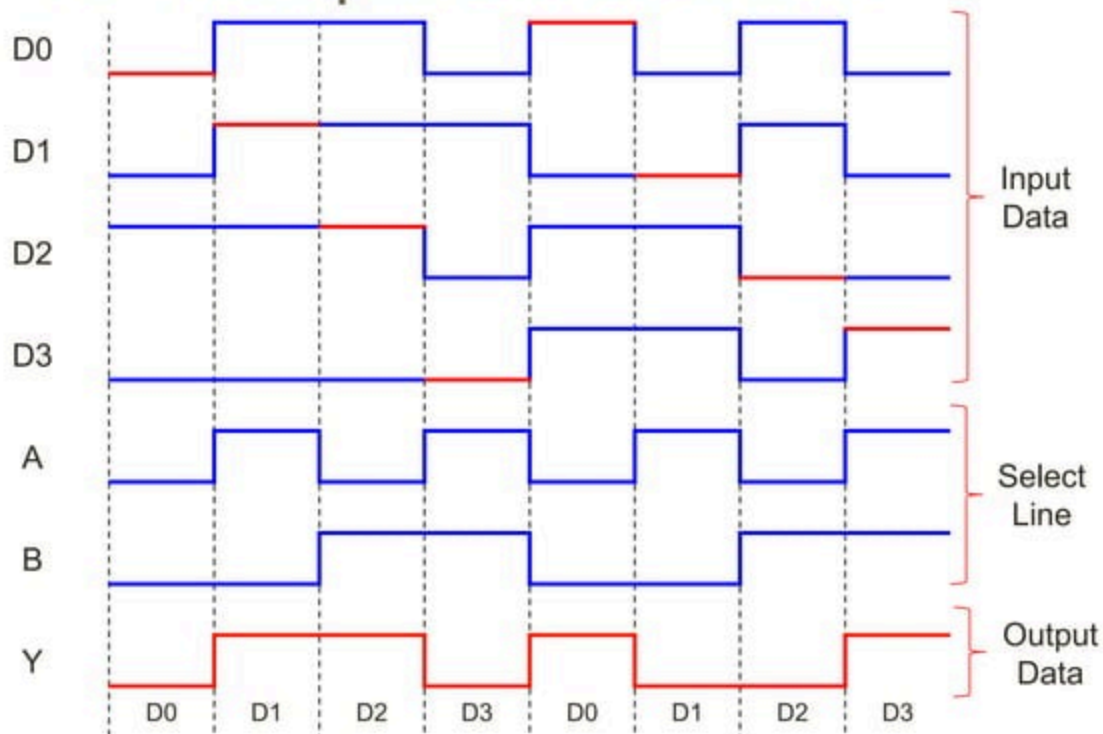
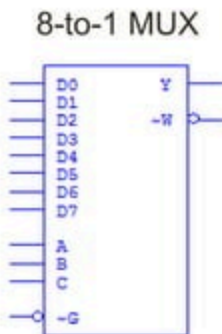
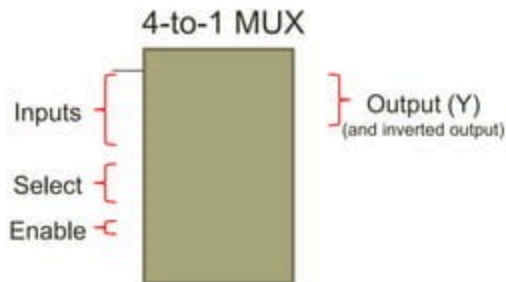


Fig. 6.2 A 4:1 Multiplexer with Strobe Input Using NAND Gates

4-to-1 Multiplexer Waveforms



Medium Scale Integration MUX



$$Y = A'B'C'D_0 + A'B'CD_1 + A'B'C'D_2$$

Combinational Logic Design Using Multiplexers

Use of multiplexers offers the following advantages:

1. Simplification of logic expression is not required.
2. It minimises the IC package count.
3. Logic design is simplified.

Available Multiplexer ICs

IC No.	Description	Output
74157	Quad 2:1 Multiplexer	Same as input
74158	Quad 2:1 Multiplexer	Inverted input
74153	Dual 4:1 Multiplexer	Same as input
74352	Dual 4:1 Multiplexer	Inverted input
74151	8:1 Multiplexer	Complementary outputs
74152	8:1 Multiplexer	Inverted input
74150	16:1 Multiplexer	Inverted input

Implement of logic expression using MUX

- For using the multiplexer as a logic element, either the truth table or one of the standard forms of logic expression must be available.

The design procedure is given below:

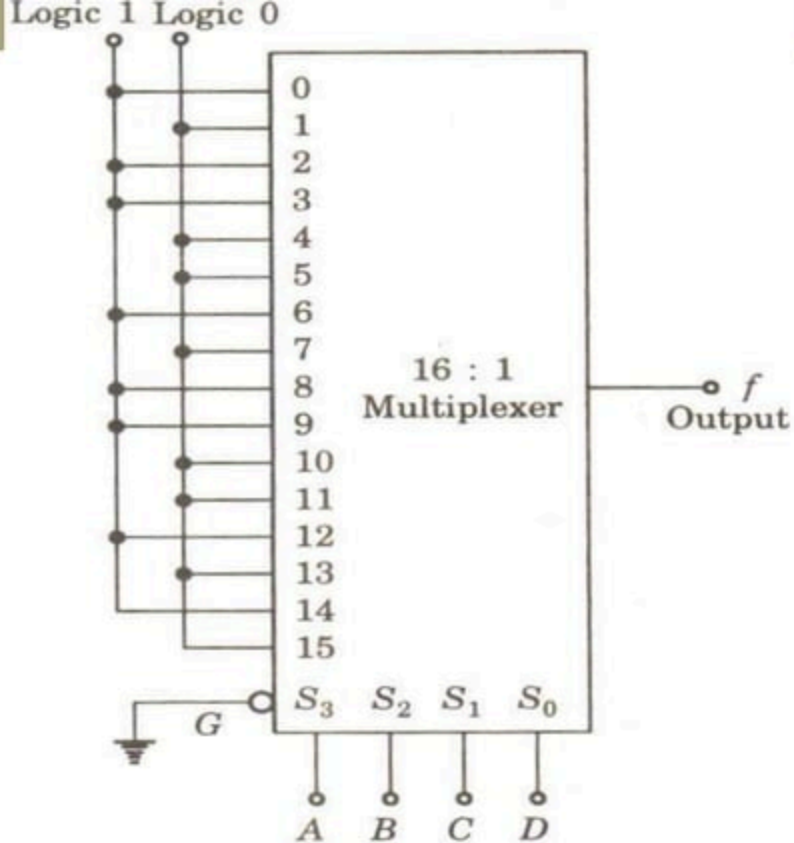
- 1. Identify the decimal number corresponding to each minterm in the expression.
- The input lines corresponding to these numbers are to be connected to logic 1 level.
- 2. All other input lines are to be connected to logic 0 level.
- 3. The inputs are to be applied to select lines.

Implement the expression using a multiplexer.

$$f(A, B, C, D) = \sum m(0, 2, 3, 6, 8, 9, 12, 14)$$

Since there are four variables, therefore, a multiplexer with four select inputs is required. The circuit of 16:1 multiplexer connected to implement the above expression

All mean terms are connected with logic 1 and remaining inputs are connected with logic 0.



Implement the expression using a multiplexer.
 $f(A, B, C, D) = \sum m(0, 2, 3, 6, 8, 9, 12, 14)$

Example 6.2

Realise the logic function of the truth table given in Table 6.3.

Table 6.3 *Truth table of Example 6.2*

Inputs				Output <i>Y</i>
<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

Solution

- (i) *First Method:* This can be realised using the method used in Example 6.1. Here, the input lines 2, 4, 6, 7, 9, 10, 11, 12, and 15 are to be connected to logic 1 and the input lines 0, 1, 3, 5, 8, 13, and 14 are to be connected to logic 0.
- (ii) *Second Method:* A four variable truth table or logic expression can be realised by using an 8:1 multiplexer instead of a 16:1 multiplexer. For this, partition the truth table as shown by dotted lines. Here the inputs A , B , and C are to be connected to S_2 , S_1 and S_0 select inputs respectively. Now, we observe the relationship between input D and output Y for each group of two rows. There are four possible values of Y and these are 0, 1, D , and $\bar{D}$. These are given in Table 6.4. From this table, we note the output Y for each of the combinations of A , B , and C , and then make the connections accordingly. The implementation of this function using an 8:1 multiplexer is shown in Fig. 6.4.

Table 6.4

Inputs			Output Y
A	B	C	
0	0	0	0
0	0	1	$\bar{D}$
0	1	0	$\bar{D}$
0	1	1	1
1	0	0	D
1	0	1	1
1	1	0	$\bar{D}$
1	1	1	D

The second method can also be used if the logic expression is specified.

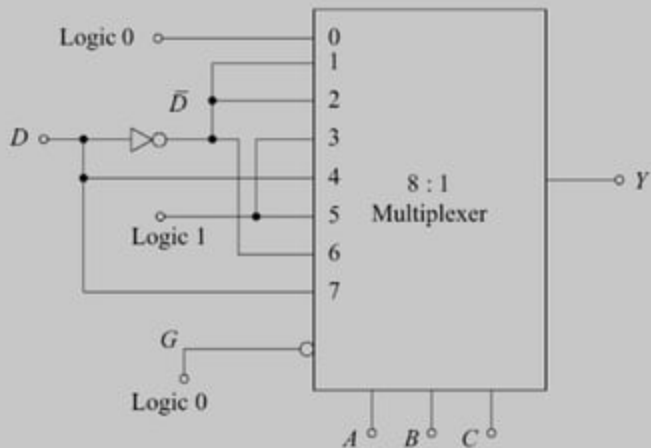


Fig. 6.4 Realisation of 4-variable Truth Table Using 8:1 Multiplexer

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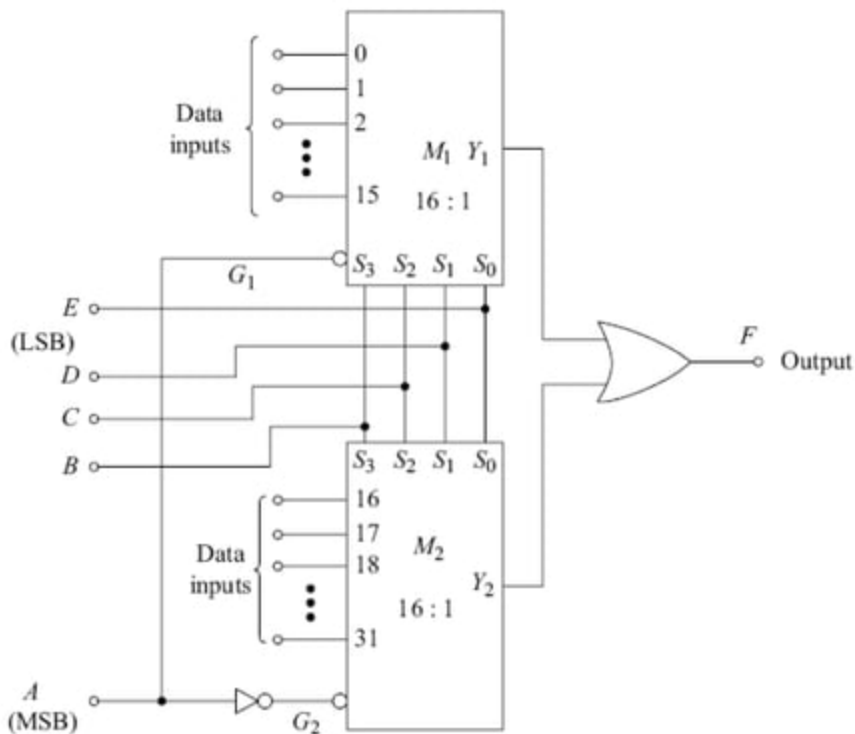


Fig. 6.5 32:1 Multiplexer Using Two 16:1 Multiplexers and One OR Gate

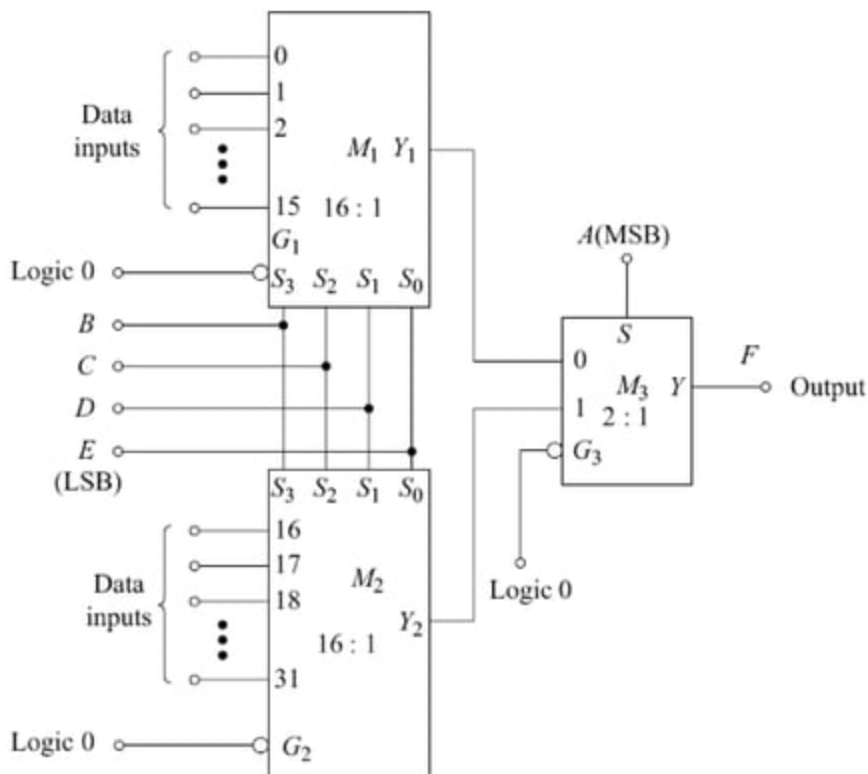
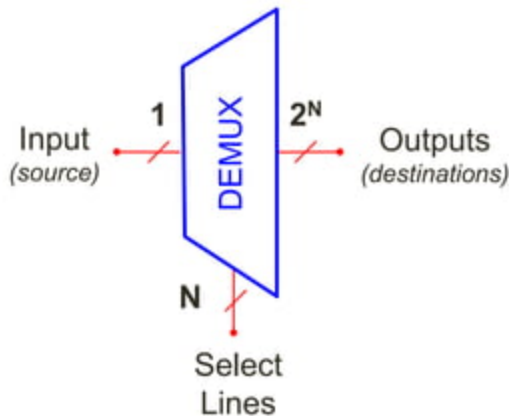


Fig. 6.6 32:1 Multiplexer Using Two 16:1 Multiplexers and One 2:1 Multiplexer

Demultiplexer (DEMUX)

- A DEMUX is a digital switch with a single input (source) and a multiple outputs (destinations).
- The select lines determine which output the input is connected to.
- DEMUX Types
 - 1-to-2 (1 select line)
 - 1-to-4 (2 select lines)
 - 1-to-8 (3 select lines)
 - 1-to-16 (4 select lines)

Demultiplexer
Block Diagram



DEMULTIPLEXERS/DECODERS AND THEIR USE IN COMBINATIONAL LOGIC DESIGN

Demultiplexer

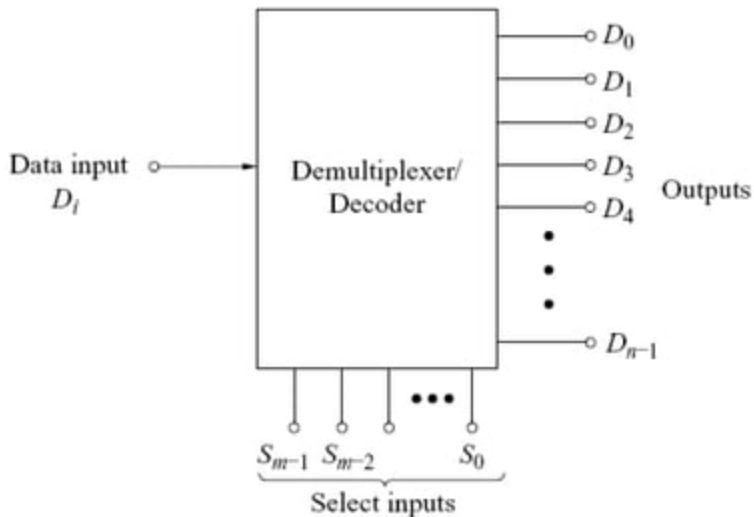
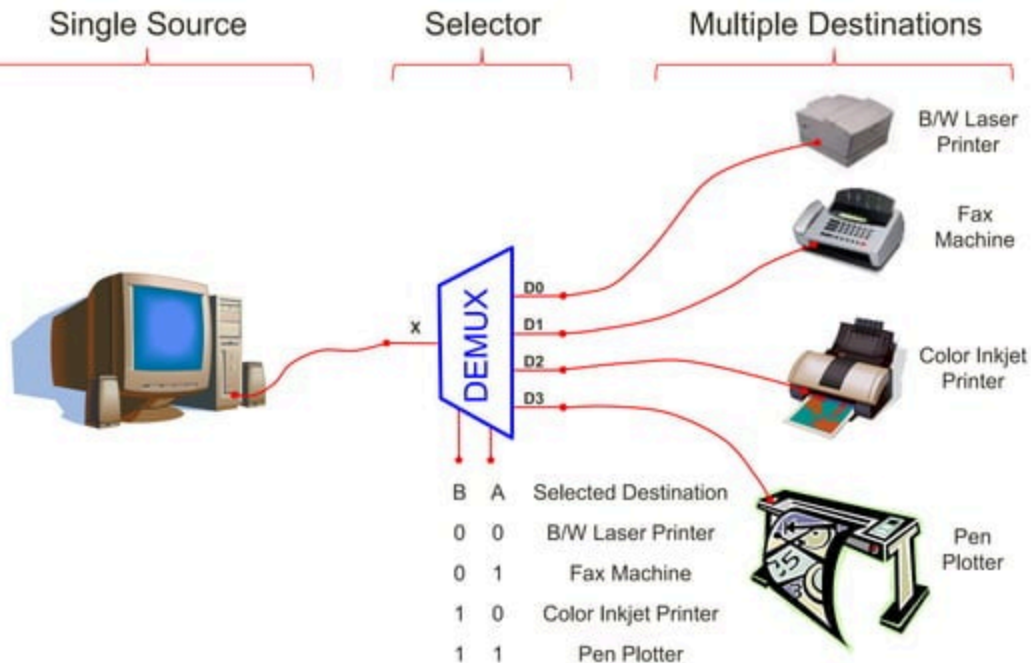
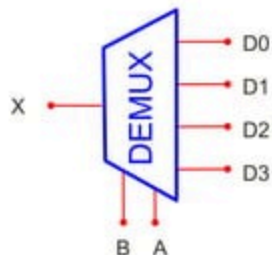
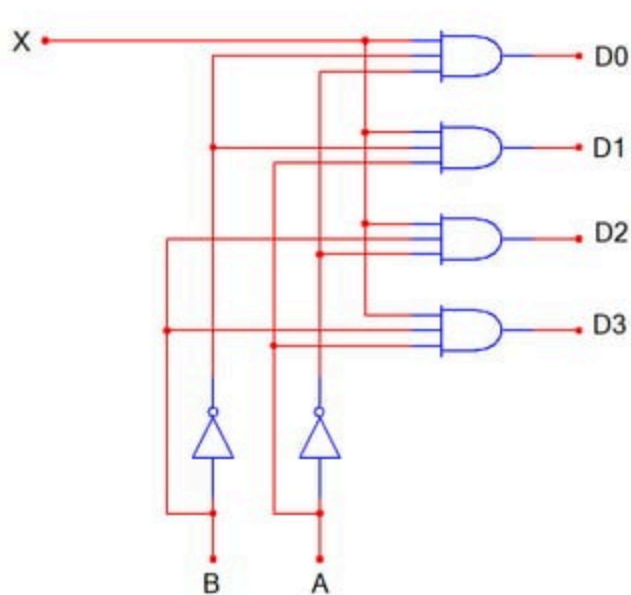


Fig. 6.7 Block Diagram of a Demultiplexer.

Typical Application of a DEMUX

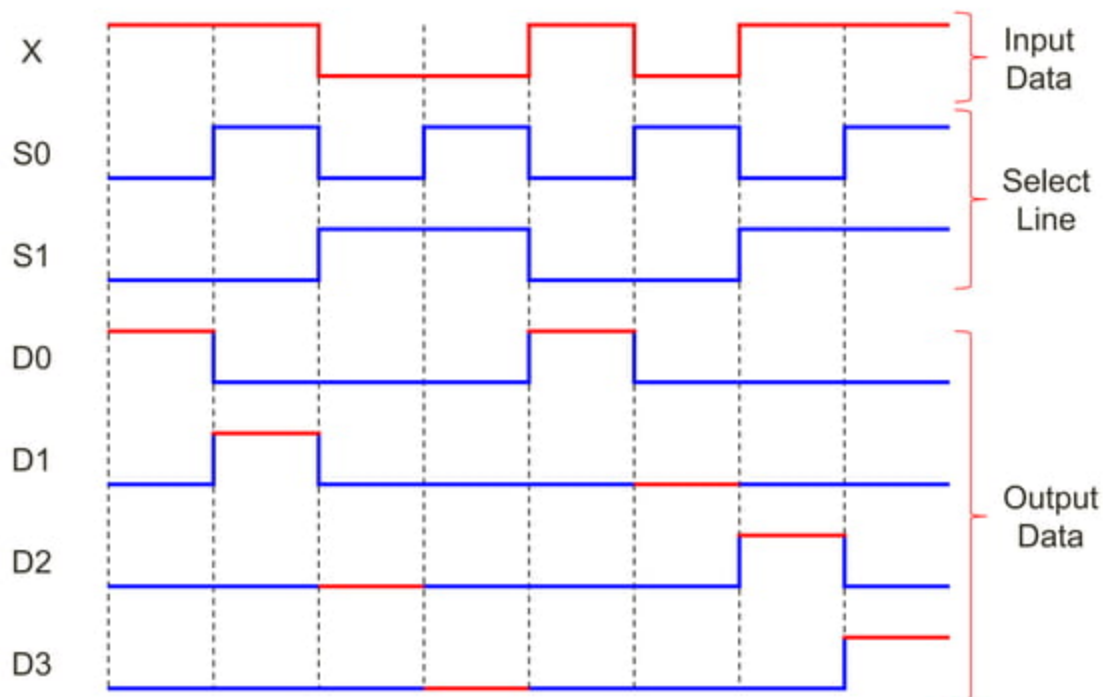


1-to-4 De-Multiplexer (DEMUX)



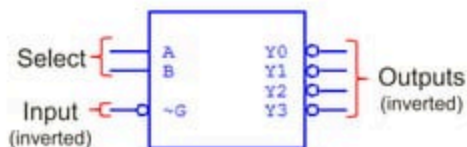
B	A	D0	D1	D2	D3
0	0	X	0	0	0
0	1	0	X	0	0
1	0	0	0	X	0
1	1	0	0	0	X

1-to-4 De-Multiplexer Waveforms

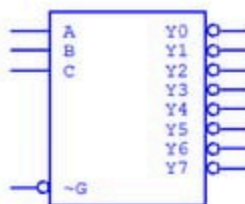


Medium Scale Integration DEMUX

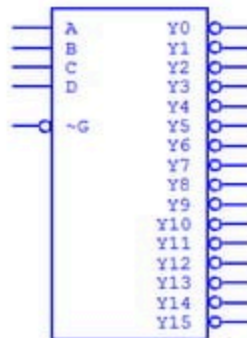
1-to-4 DEMUX



1-to-8 DEMUX



16-to-1 MUX



Most Medium Scale Integrated (MSI) DEMUXs have inverted outputs. This is done because it requires few logic gates to implement DEMUXs with inverted outputs rather than no-inverted outputs.

Available Demultiplexer ICs

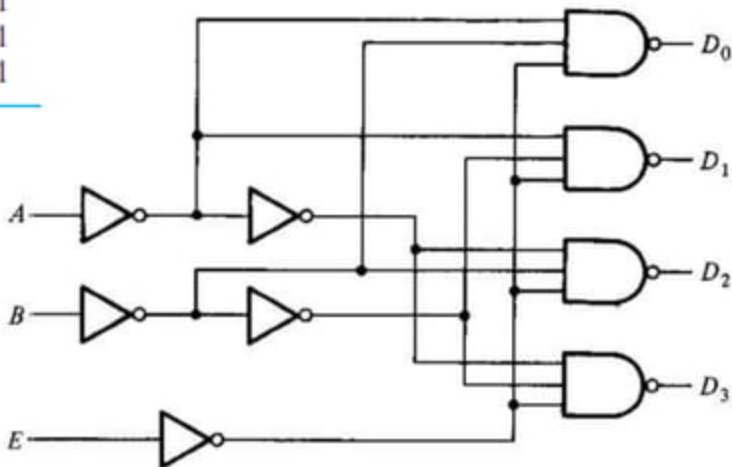
IC No.	Description	Output
74139	Dual 1:4 Demultiplexer (2-line-to-4-line decoder)	Inverted input
74155	Dual 1:4 Demultiplexer (2-line-to-4-line decoder)	1 <i>Y</i> –Inverted input 2 <i>Y</i> –Same as input
74156	-do-	Open-collector 1 <i>Y</i> –Inverted input 2 <i>Y</i> –Same as input
74138	1:8 Demultiplexer (3-line-to-8-line decoder)	Inverted input
74154	1:16 Demultiplexer (4-line-to-16-line decoder)	Same as input
74159	-do-	Same as input Open-collector

DECODERS

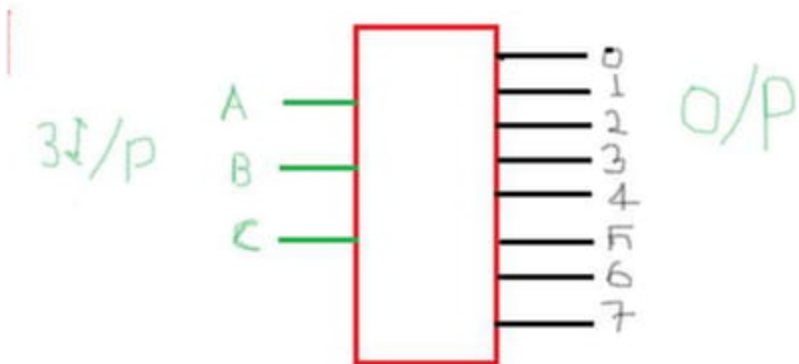
- Decoder is a combinational circuit that converts binary information from “n” input lines to a maximum of 2^n unique output lines . if the n-bit coded information has unused combinations ,the decoder may have fewer than 2^n outputs.

2:4 Decoder

<i>E</i>	<i>A</i>	<i>B</i>	<i>D</i> ₀	<i>D</i> ₁	<i>D</i> ₂	<i>D</i> ₃
1	<i>X</i>	<i>X</i>	1	1	1	1
0	0	0	0	1	1	1
0	0	1	1	0	1	1
0	1	0	1	1	0	1
0	1	1	1	1	1	1



3-to-8-line decoder



3:8 Line Decoder

TABLE 10-1 3:8 LINE DECODER

Inputs			Outputs							
x	y	z	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

Circuit diagram

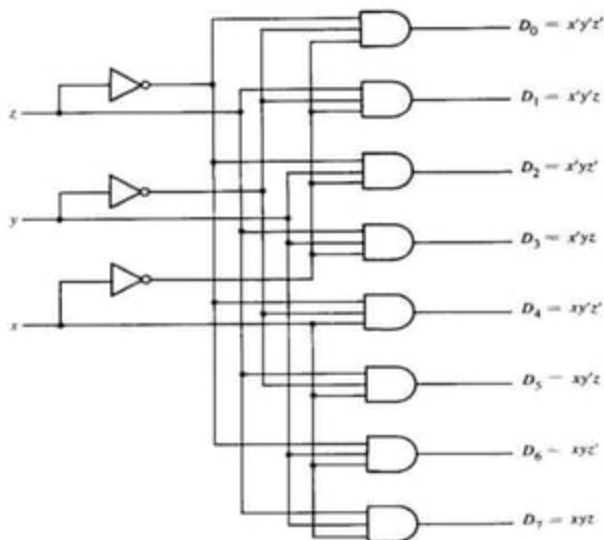
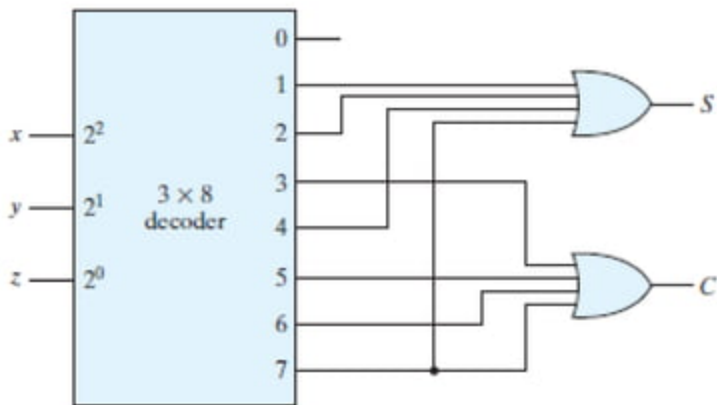


FIGURE 5-8
A 3-to-8 line decoder

Full Adder using Decoder

$$S(x, y, z) = \Sigma(1, 2, 4, 7)$$

$$C(x, y, z) = \Sigma(3, 5, 6, 7)$$



Implement the following multi-output combinational logic circuit using a 4-to-16-line decoder.

$$F1 = \sum m(1, 2, 4, 7, 8, 11, 12, 13)$$

$$F2 = \sum m(2, 3, 9, 11)$$

$$F3 = \sum m(10, 12, 13, 14)$$

$$F4 = \sum m(2, 4, 8)$$

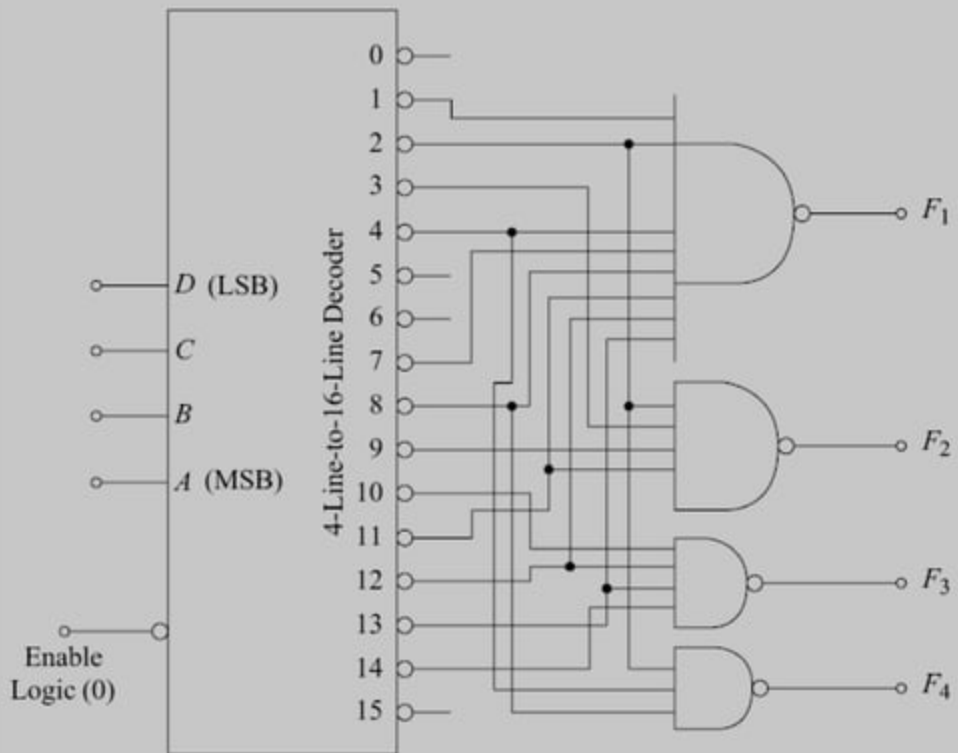


Fig. 6.8 Implementation of Combinational Logic Circuit of Ex. 6.3

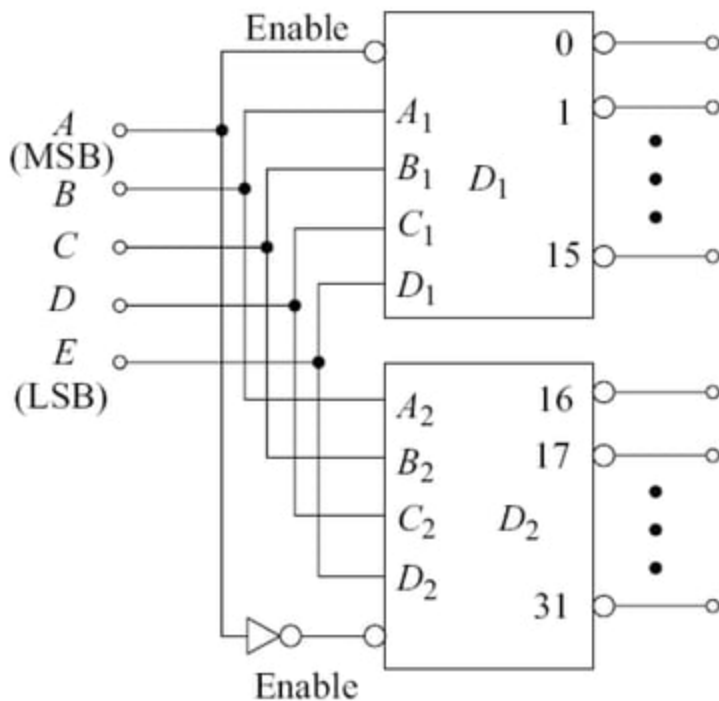


Fig. 6.9 *5-Line-to-32-Line Decoder Using Two*
 ----- *4-line-to-16-Line Decoders*

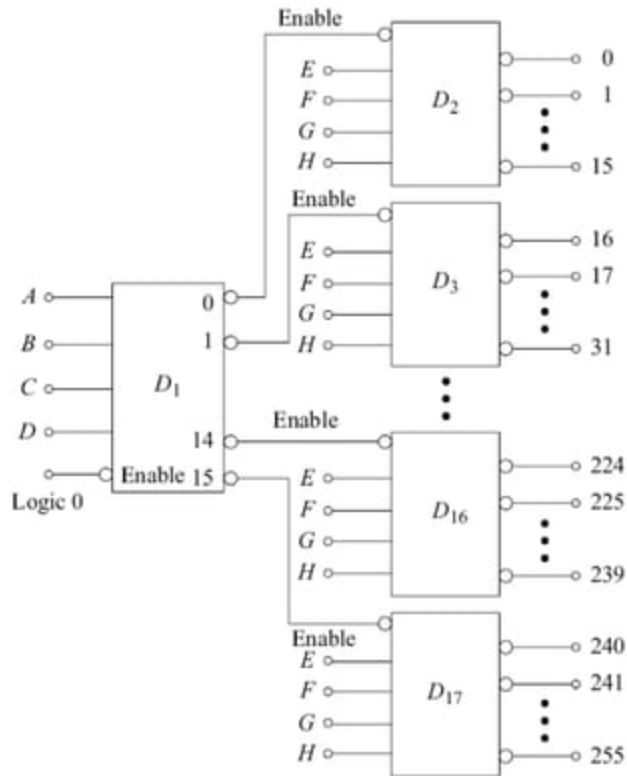


Fig. 6.10 8-Line-to-256-Line Decoder Using 4-Line-to-16-Line Decoders Tree

Thankyou